

Lie Theory for the Impatient

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Abstract

quaternionic

This document collects a variety of definitions and results from Lie theory.

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1 Topics in Representation Theory

1.1 Miniscule Weights

Definition 1 (miniscule). A dominant integral weight ω is *miniscule* if and only if $(\omega, \beta) \leq 1$ for all positive coroots β .

Remark 2. Nonzero miniscule weights are fundamental: $(\omega, \alpha_i^\vee) \leq 1$, with equality for at most one i .

Proposition 3. Let ω be a dominant integral weight. The following are equivalent.

- (i) ω is miniscule.
- (ii) All weights of L_ω belong to $W\omega$.
- (iii) If λ is dominant integral, and $\omega - \lambda \in Q^+$, then $\lambda = \omega$.

Sketch. For (iii) implies (ii), note a weight λ of L_ω has $w\lambda$ dominant for some w , but then (iii) means $w\lambda = \omega$. For (ii) implies (i), suppose (i) is false, and we hunt for a suitable weight space in L_ω . Well, choose a root α with $(\omega, \alpha) > 1$, and check that $\omega - \alpha$ is a weight space of L_ω not in $W\omega$.

It remains for (i) implies (iii), which is some difficult combinatorics that I do not understand. ■

Proposition 4. Suppose ω is miniscule, and λ is dominant integral. Then

$$L_\omega \otimes L_\lambda = \bigoplus_{\substack{\gamma \in W\omega \\ \lambda + \gamma \geq 0}} L_{\lambda + \gamma}.$$

Sketch. Use the Weyl character formula to expand both sides. ■

Proposition 5. Each coset in P/Q admits a unique miniscule weight.

Proof. In one direction, suppose ω is the dominant integral weight in a coset of P/Q of minimal height. Then the only allowed weights of L_ω are in $W\omega$ for height reasons, so ω is miniscule.

In the other direction, if ω and ω' are miniscule weights in the same coset of P/Q , we know $\omega_1 - \omega_2 \in Q$. To show $\omega_1 = \omega_2$, note that $(\omega_1 - \omega_2, \beta) \leq 1$ for each coroot β : we may assume β is positive, and then (ω_1, β) and (ω_2, β) are both in $\{0, 1\}$. But if $\omega \in Q$ has $(\omega, \beta) \leq 1$ for all coroots β , we must have $\omega = 0$: write $\omega = \sum_i m_i \alpha_i$ as the minimal counterexample, but one can show that $s_j \omega = \omega - \alpha_j$ is a smaller counterexample for some j (choose j so that m_j and $(\omega, \alpha_j^\vee)$ have the same sign, which we may assume is positive). ■

1.2 Representations of GL_n

Theorem 6 (Schur–Weyl duality). Let V be a complex vector space of dimension n . Then there is a decomposition

$$V^{\otimes N} = \bigoplus_{\substack{\lambda = (\lambda_1, \dots, \lambda_n) \\ \lambda \vdash N}} L_\lambda \otimes \pi_\lambda$$

into irreducible representations of $GL_n(\mathbb{C}) \times S_N$, where L_λ is the irreducible representation of $GL(V)$ of highest weight λ . If $n \geq N$, the the set $\{\pi_\lambda\}_\lambda$ exhausts the set

Sketch. One uses the double-centralizer lemma on the following subalgebras of $\text{End}_{\mathbb{C}} V^{\otimes N}$: define A to be the image of $U\mathfrak{gl}_n(\mathbb{C})$, and define B to be the image of $\mathbb{C}[S_N]$. For the last assertion, one notes that the quotient map $\mathbb{C}[S_N] \rightarrow B$ is an isomorphism. ■

Definition 7. For a partition $\lambda \vdash N$ and a vector space V over $\mathbb{C}$, define the *Schur functor*

$$S^\lambda V := \text{Hom}_{S_N}(\pi_\lambda, V^{\otimes N}).$$

Theorem 8 (Invariant theory). Fix a finite-dimensional vector space V . Then we can describe a spanning set of $\text{GL}(V)$ -invariant functions on the space

$$\bigotimes_i (V^{\otimes m_i} \otimes V^{*\otimes n_i})^{\otimes d_i}$$

as follows: start with d_i vertices for each i , required to have m_i incoming and n_i outgoing arrows. For each graph Γ built from such vertices, define F_Γ as the associated convolution. ■

Sketch. We are hunting for $\text{GL}(V)$ -invariant functionals on the given tensor power. By rearranging, this should amount to calculating $\text{GL}(V)$ -invariant endomorphisms of $V^{\otimes N}$ for some N , which are spanned by permutations by Schur–Weyl duality. ■

Definition 9. Fix a partition λ with at most n parts. By the Weyl character formula, the character of the irreducible representation L_λ of $\text{GL}_n(\mathbb{C})$ is given by the *Schur polynomial*

$$s_\lambda(x_1, \dots, x_n) = \frac{\det [x_i^{\lambda_j + n - j}]_{ij}}{\prod_{i < j} (x_i - x_j)}.$$

Theorem 10 (Frobenius character formula). Suppose $\sigma \in S_N$ is of cycle type $(\ell_1, \dots, \ell_r)$. Then for a partition λ of N , the character value $\text{tr}(\sigma; \pi_\lambda)$ is the coefficient of $x_1^{\lambda_1 + n - 1} \dots x_n^{\lambda_n}$ of the polynomial

$$\prod_{i < j} (x_i - x_j) \cdot \prod_{i=1}^r (x_1^{\ell_i} + \dots + x_n^{\ell_i}).$$

Sketch. For diagonal $x \in \text{GL}_n(\mathbb{C})$, one can compute the trace of $x\sigma$ acting on $V^{\otimes N}$ in two ways: direct calculation on the standard basis of $V^{\otimes N}$ shows that

$$\text{tr}(x\sigma; V^{\otimes N}) = \prod_{i=1}^r (x_1^{\ell_i} + \dots + x_n^{\ell_i}).$$

Alternatively, we can compute the trace of $x\sigma$ in the decomposition of $V^{\otimes N}$ given by Schur–Weyl duality. ■

Theorem 11 (Howe duality). For vector spaces V and W , we have

$$S^N(V \otimes W) = \bigoplus_{\lambda \vdash N} S^\lambda V \otimes S^\lambda W.$$

Sketch. The key idea is that $S^N(V \otimes W) = (V^{\otimes N} \otimes W^{\otimes N})^{S_N}$. One now expands $S^{\otimes N}$ and $W^{\otimes N}$ using Schur–Weyl duality. Eventually, one needs to know that $(\pi_\lambda \otimes \pi_\mu)^{S_N}$ is at most one-dimensional (by Schur’s lemma), and it is one-dimensional if and only if $\lambda = \mu$ (because π_μ is self-dual because its character is real-valued). ■

Proposition 12. Let λ be a partition of N . Then

$$\mathbb{C}[S_{N+1}] \otimes_{\mathbb{C}[S_N]} \pi_\lambda = \bigoplus_{\mu \in \lambda + \square} \pi_\mu.$$

Sketch. It is enough to check that both representations give the same multiplicity space in $V^{\otimes(N+1)}$ for V of large dimension. The right-hand side gives $V \otimes S^\lambda V$ because V is miniscule of weight $(1, 0, \dots, 0)$. The left-hand side gives $V \otimes S^\lambda V$ by applying Frobenius reciprocity and then Schur–Weyl duality. ■

Corollary 13. Let λ be a partition. Then $\pi_\lambda \otimes \text{sgn} = \pi_{\lambda^\dagger}$.

Sketch. Induct on $|\lambda|$. Then the proposition implies

$$\bigoplus_{\mu \in \lambda + \square} \pi_\mu \otimes \text{sgn} = \bigoplus_{\mu \in \lambda^\dagger + \square} \pi_\mu.$$

One now uses the Casimir element of $\text{GL}_n(\mathbb{C})$ (equivalently, the Jucys–Murphy element of $\mathbb{C}[S_{N+1}]$ given by $\sum_{i < j} (i, j)$) to separate out the representations appropriately by eigenvalue. ■

1.3 The Automorphism Group

Definition 14 (adjoint group). For a finite-dimensional Lie algebra $\mathfrak{g}$, let G_{ad} be $\text{Aut}(\mathfrak{g})^\circ$.

Remark 15. A computation on differentials shows $\text{Lie } G_{\text{ad}} = \text{Der } \mathfrak{g}$. If $\mathfrak{g}$ is semisimple, then the natural map $\mathfrak{g} \rightarrow \text{Der } \mathfrak{g}$ is an isomorphism, so $\text{Lie } G_{\text{ad}} = \mathfrak{g}$.

Lemma 16. Let $\mathfrak{h}$ be a Cartan subalgebra, and let $T \subseteq G_{\text{ad}}$ be the corresponding subgroup. If $h \in \text{Aut}(\mathfrak{g})$ fixes $\mathfrak{h}$ and the root spaces, then $h \in T$.

Sketch. For each simple root α_i , find $c_i \in \mathbb{C}$ such that the action of h on $\mathfrak{g}[\alpha_i]$ is $\exp(c_i)$. Then the automorphism h of $\mathfrak{g}$ equals the automorphism $\exp(\sum_i c_i \omega_i^\vee)$. ■

Lemma 17. Let $\mathfrak{h}$ be a Cartan subalgebra, and let $T \subseteq G_{\text{ad}}$ be the corresponding subgroup. If $g \in G_{\text{ad}}$ preserves $\mathfrak{h}$ and a set of simple roots, then $g = 1$.

Sketch. The collection of $g \in G_{\text{ad}}$ which preserve $\mathfrak{h}$ and a set of simple roots must then permute the simple root spaces and hence induces an automorphism σ_g of the Dynkin diagram. The idea is that this automorphism of the Dynkin diagram should be reflected in the fundamental representations, which are discrete, so $g = 1$ will be forced by G_{ad} being connected.

In a few more words, note $g \in G_{\text{ad}}$ acts on $\text{Rep } \mathfrak{g}$ by sending ρ to $\rho \circ \text{Ad}_{g^{-1}}$.

- On one hand, if g preserves $\mathfrak{h}$ and a set of simple roots (so that it preserves $\mathfrak{n}_+$), we see that g preserves singular vectors, so for any dominant integral λ , we see $\rho_\lambda \circ \text{Ad}_{g^{-1}} = \rho_{\sigma\lambda}$.

- On the other hand, this action is essentially trivial: by the Weyl character formula, for any dominant integral weight λ , we see $g \mapsto \text{tr}(-; \rho_\lambda \circ \text{Ad}_{g^{-1}})$ is a continuous family of polynomials valued in the characters. But the characters are a discrete set of polynomials, so this continuous family is constant. We conclude $\rho_\lambda \circ \text{Ad}_{g^{-1}} = \rho_\lambda$. ■

Proposition 18. Let $\mathfrak{h}$ be a Cartan subalgebra, and let $T \subseteq G_{\text{ad}}$ be the corresponding subgroup. Then $N(T)/T \cong W$.

Sketch. The main point is to construct lifts of the simple reflections of W . Accordingly, for each simple root α_i , let $\dot{s}_i \in G_{\text{ad}}$ be the image of $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ along the map $\text{SL}_2(\mathbb{C}) \rightarrow G_{\text{ad}}$ defined by lifting $\mathfrak{sl}_{2i} \subseteq \mathfrak{g}$. Then $\text{Ad}_{\dot{s}_i}$ sends $\alpha_i \mapsto -\alpha_i$ and fixes $\{\alpha_i\}^\perp$ (because the full one-parameter subgroup $t \mapsto \exp(th_i)$ does), so the action agrees with the action of s_i on $\mathfrak{h}$. Thus, $\dot{s}_i$ stabilizes $\mathfrak{h}$ and lives in $N(T)$.

Extending this construction multiplicatively produces a subgroup $N \subseteq N(T)$ such that $N/T \cong W$. To see $N = N(T)$, note that any $g \in N(T)$ preserves $\mathfrak{h}$, and we may adjust g by $N \rightarrow W$ so that g preserves a set of simple roots of $\mathfrak{h}$, which forces $g = 1$. ■

Proposition 19. Let D be the Dynkin diagram, acting on $\mathfrak{g}$ by permuting the Serre presentation. Then the natural map

$$\text{Aut } D \rightarrow \pi_0 \text{Aut}(\mathfrak{g})$$

is an isomorphism.

Sketch. We already know $G_{\text{ad}} \cap \text{Aut } D$ is trivial, so the difficulty is surjectivity. Well, for $g \in \text{Aut}(\mathfrak{g})$, choose a g -stable Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$. Adjusting by W , we may assume g preserves a set of simple roots. Adjusting by $\text{Aut } D$, we may assume g preserves the set of simple roots pointwise. Adjusting by H , we may assume g preserves the root spaces pointwise. It follows $g = 1$. ■

1.4 Miscellaneous Topics

Proposition 20. Let w_0 be the long Weyl element. For any dominant integral weight λ , we have $L_\lambda^* = L_{-w_0\lambda}$.

Sketch. The lowest weight of L_λ is $w_0\lambda$, which we see by considering the Weyl orbit. Negate to get a dominant weight. ■

Definition 21 (maximal root). The *maximal root* θ is the dominant integral weight associated to the adjoint representation, which is a root by the root decomposition.

Remark 22. Because L_θ is a quotient of the Verma module M_θ , we see that any root α has $\theta - \alpha \in Q^+$. It follows that the height of θ is maximal among roots.

Remark 23. The maximal root is always long, easily seen from the classification.

Remark 24. The maximal root is a fundamental weight outside of types A and C , which is checked from the classification.

Definition 25 (Coxeter number). The Coxeter number is $(\theta, \rho^\vee) + 1$. This is one more than the height of the maximal root. The dual Coxeter number is $(\tilde{\theta}^\vee, \rho) + 1$, where $\tilde{\theta}^\vee$ is the coroot of θ .

Remark 26. If $\mathfrak{g}$ is simply laced, the dual root system agrees with the root system, so $\tilde{\theta}^\vee = \theta^\vee$, so the Coxeter number and dual Coxeter number are equal. This is false without the simply laced assumption.

Definition 27 (principle $\mathfrak{sl}_2$ -triple). The principal $\mathfrak{sl}_2$ -triple (e, f, h) has $e = \sum_i e_i$ and $h = 2\rho^\vee$ and $f = \sum_i (2\rho^\vee, \omega_i) f_i$.

Definition 28 (exponents). Consider $\mathfrak{g}$ as a representation of the principle $\mathfrak{sl}_2$. For $m \geq 0$, define $r_m := \dim \mathfrak{g}[2m]$, which is the number of (positive) roots of height m if $m > 0$. Then the exponents are the positive integers m for which $r_m > r_{m+1}$, and the multiplicity is $r_m - r_{m+1}$.

Example 29. We have $r_0 = \text{rank } \mathfrak{g}$ by the root decomposition, $r_1 = \text{rank } \mathfrak{g}$ is the number of simple roots, and $r_2 = \text{rank } \mathfrak{g} - 1$ because $\alpha_i + \alpha_j$ is a root if and only if α_i and α_j are adjacent in the Dynkin diagram (seen from the classification of rank-2 root systems).

Definition 30 (type). Let V be a finite-dimensional representation. Then V is

- of complex type if and only if $V \not\cong V^*$,
- of real type if and only if $V \cong V^*$ by a symmetric bilinear form, and
- of quaternionic type if and only if $V \cong V^*$ by an alternating bilinear form.

Proposition 31. Let λ be a dominant integral weight. Then

- is of complex type if and only if $\lambda \neq -w_0\lambda$,
- is of real type if and only if $\lambda = -w_0\lambda$ and $(2\rho^\vee, \lambda)$ is even, and
- of quaternionic type if and only if $\lambda = -w_0\lambda$ and $(2\rho^\vee, \lambda)$ is odd.

Sketch. Here, (a) follows because $L_\lambda^* = L_{-w_0\lambda}$. One checks (b) and (c) by passing to the principal $\mathfrak{sl}_2$ -triple, which features a highest weight vector v_λ of multiplicity one by height considerations. ■

2 The Peter–Weyl Theorem

2.1 Measure Theory

Theorem 32 (Haar). Let G be a locally compact group. Then G admits a unique (up to scalar) left-invariant Radon measure μ . There is an analogous statement for right-invariance.

Sketch. We ignore uniqueness. Roughly speaking, it follows from the Fubini–Tonelli theorem, which we see by comparing double integrals over μ and ν .

The difficulty lies in existence. We only discuss some special cases.

- If G is discrete, then the counting measure suffices.

- Suppose G is a Lie group (or more generally the topological space associated to the rational points of an algebraic group over a local field). If G is a Lie group, one can construct measures by choosing a global section ξ of $\det TG$. To have left-invariance, we must have $\xi_g = L_g^* \xi_1$, so it is enough to choose $\xi_1 \in \det T_e G$.
- Suppose G is compact and separable. Let $\{g_i\}$ be a dense subsequence, and choose positive weights $\{p_i\}$ so that $\sum_i p_i = 1$. It is enough to exhibit an invariant positive continuous functional I on $C(G; \mathbb{R})$. The idea is to view I as the “average” of a function, which we can compute iteratively with the approximation

$$Af(x) := \sum_i p_i f(xg_i).$$

In particular, A is a bounded linear operator of norm one; also, setting $v(f) := \max f - \min f$ to be the variation, if $v(f) > 0$, one has that $\max Af < \max f$ and $\min Af > \min f$ and so $v(Af) < v(f)$. One can use the Arzelà–Ascoli theorem to show that $\{A^n f\}_n$ admits a convergent subsequence, and we see that the limit must be constant, which one takes to be $I(f)$. (In fact, $\{A^n f\}_n$ converges: the sequences $\{\max A^n f\}_n$ and $\{\min A^n f\}_n$ are monotone, bounded, and they have a subsequence converging to $I(f)$.) ■

Definition 33 (unimodular). A locally compact group G is *unimodular* if and only if left-invariant Radon measure on G are also right-invariant.

Proposition 34. Fix a (real) Lie group G . Then G is unimodular if and only if the character $|\det \text{Ad}|$ is trivial.

Sketch. Choose nonzero $\xi \in \det \mathfrak{g}$. We want to know if the left-invariant global section $g \mapsto |L_g^* \xi|$ is also right-invariant. It is equivalent to ask if $g \mapsto |R_{g^{-1}}^* L_g \xi|$ is trivial, which is equivalent to ask if $|\xi|$ is conjugation-invariant. ■

Example 35. If G is compact, then G is unimodular: any continuous character $G \rightarrow \mathbb{R}^+$ has compact and hence trivial image.

Definition 36. A Lie algebra $\mathfrak{g}$ is unimodular if and only if $\det \text{ad}$ is the trivial representation.

Remark 37. If G is connected, then the character $\det \text{Ad}: G \rightarrow \mathbb{R}^\times$ has connected image and hence already outputs to $\mathbb{R}^+$. Thus, G is unimodular if and only if $\mathfrak{g}$ is unimodular.

Example 38. If $\mathfrak{g}$ is abelian, it is unimodular because ad is zero. If $\mathfrak{g}$ is semisimple, then $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, so $\mathfrak{g}$ is unimodular because any character $\mathfrak{g} \rightarrow \mathbb{C}$ is zero. Taking the sum shows that $\mathfrak{g}$ is unimodular if $\mathfrak{g}$ is reductive.

2.2 Matrix Coefficients

Proposition 39. Let G be a compact group. Then any finite-dimensional representation V of G is unitary.

Sketch. Choose any Hermitian form on V , and then average it over G using integration to produce a G -invariant Hermitian form. ■

Proposition 40. Let G be a Lie group. For any continuous finite-dimensional representation V , the matrix coefficients of V are smooth.

Sketch. Let V^{sm} be the subspace of V consisting of $v \in V$ for which any matrix coefficient $g \mapsto \ell(gv)$ is smooth. Because convolution picks up the best regularity properties, we know that $\varphi \cdot v$ is smooth for any $v \in V$ and $\varphi \in C^\infty(G)$. By using an approximate identity, one finds that V^{sm} is dense in V . Thus, $V^{\text{sm}} = V$ because V is finite-dimensional. ■

Proposition 41 (Orthogonality). Let G be a compact group, and let V and W be non-isomorphic irreducible finite-dimensional representations. For $v \in V$ and $v^* \in V^*$ and $w \in W$ and $w^* \in W^*$, we have

$$\int_G \langle gv, v^* \rangle \overline{\langle gw, w^* \rangle} dg = 0.$$

Sketch. Because W is unitary, $\overline{W} = W^*$. Thus, it is enough to show

$$\int_G \langle gv, v^* \rangle \langle w, gw^* \rangle dg = 0.$$

under the assumption $V \not\cong W^*$. Well, fix v^* and w , and then the associated map $V \otimes W \rightarrow \mathbb{C}$ is G -invariant and hence vanishing unless $V \cong W^*$. ■

Proposition 42 (Orthogonality). Let G be a compact group, and let V be an irreducible finite-dimensional representation. For $v, w \in V$ and $v^*, w^* \in V^*$, we have

$$\int_G \langle gv, v^* \rangle \overline{\langle gw, w^* \rangle} dg = \frac{1}{\dim V} \langle v, w^* \rangle \langle w, v^* \rangle.$$

Sketch. Upon identifying $\overline{V}$ with V^* , the left-hand side is

$$\int_G \langle gv, v^* \rangle \langle w, gw^* \rangle dg.$$

If one fixes v^* and w , this is an invariant Hermitian form in v and w^* , which is unique up to scalar, which explains the factor of $\langle v, w^* \rangle$ on the right. The remaining scalar is still a Hermitian form in (v, w^*) , which is unique up to scalar again and explains the factor of $\langle w, v^* \rangle$ on the right. To get the remaining scalar $1/\dim V$, sum both sides over an orthonormal basis of V . ■

Definition 43 (finite). For a continuous representation V of a compact group G , let V^{fin} be the subspace of G -finite vectors, where v is G -finite if and only if $\text{span}\{gv : g \in G\}$ is finite-dimensional.

Corollary 44. Let G be a compact group. Consider the matrix coefficient map

$$\xi: \bigoplus_V (V \otimes V^*) \rightarrow L^2(G)$$

given on pure tensors by sending $v \otimes v^*$ to $g \mapsto \langle v, gv^* \rangle$. Then ξ is $(G \times G)$ -invariant, an isometric embedding, and has image given by $L^2(G)^{\text{fin}}$.

Sketch. Here, $G \times G$ acts on $C(G)$ by left and right translation, so invariance is checked directly. The isometry property follows from the orthogonality relations.

As for the image, certainly the image contains only G -finite vectors. Conversely, if $\varphi \in L^2(G)$ is G -finite, we may as well suppose that φ generates an irreducible representation V , so φ lives in a multiplicity space of $\text{Hom}_G(V, L^2(G))$. Thus, it is enough to show that the matrix coefficient map $V^* \rightarrow \text{Hom}_G(V, L^2(G))$ is an isomorphism: it turns out that any G -invariant map $V \rightarrow L^2(G)$ is uniquely determined by one value, which can then be spread out to a canonical invariant map $V \rightarrow C(G)$, and then there is an inverse map sending $\varphi: V \rightarrow L^2(G)$ to the functional $v \mapsto \varphi(v)(1)$. ■

rewrite
to show
 K -finite
implies
smooth

2.3 Functional Analysis

Definition 45 (compact). A bounded linear operator A on a Hilbert space H is *compact* if and only if it is the limit (in the norm topology) of finite-rank operators $H \rightarrow H$.

Example 46. Let M be a compact space, and choose $K \in C(M \times M)$. Then the integral operator

$$Af(y) := \int_M K(x, y)f(x) dx$$

is compact. By embedding, we may assume $M = [-1, 1]^n$. Then we may approximate the integral as a Riemann sum on a tiling by hypercubes of length $1/n$, essentially replacing K by a locally constant function (which then has finite rank).

Lemma 47. Let A be a compact operator on a Hilbert space H . If $\{v_n\}_n$ is bounded, then $\{Av_n\}_n$ admits a convergent subsequence.

Sketch. If A has finite rank, there is nothing to do because any sequence in a bounded subset of a finite-dimensional vector space admits a convergent subsequence. In general, choose $\{A_k\}_k$ converging to A , and inductively define v_n^{k+1} to be a subsequence of v_n^k (with $v_n^0 = v_n$) so that $A_{k+1}v_n^{k+1}$ converges. Then use the diagonal subsequence v_n^n as the required subsequence. ■

Theorem 48 (Hilbert–Schmidt). Let A be a self-adjoint compact operator on a Hilbert space H . Then there is an orthogonal decomposition

$$H = \ker A \oplus \widehat{\bigoplus_{\lambda} H_{\lambda}},$$

where each eigenspace H_{λ} is finite-dimensional, and the set of eigenvalues is finite or tends to zero.

Sketch. If A has finite rank, there is nothing to do. It is essentially enough to prove the theorem for A^2 (and then diagonalize everything again against A). The key difficulty is exhibiting a single eigenvector for A ; once we find one eigenvector, we can (roughly speaking) pass to its orthogonal complement and inductively downwards.

To find our eigenvector, we will actually find the largest eigenvector: we will find a vector in H with eigenvalue $\|A\|^2$. Indeed, let $\{A_n\}_n$ be finite-rank operators approaching A , and let v_n be an eigenvector of A_n^2 with eigenvalue $\|A_n\|^2$; by passing to a subsequence, we may assume $\{v_n\}$ approaches some v , so it follows $A^2v = \lim_n A_nv_n = \|A\|^2v$. Now, pass to the orthogonal complement, and our sequence of eigenvalues decreases, and it must tend to zero by compactness of A . ■

2.4 The Peter–Weyl Theorem

Theorem 49 (Peter–Weyl). Let G be a compact group. Then $L^2(G)^{\text{fin}}$ is dense in $L^2(G)$.

Sketch. Let $\{\varphi_n\}_n$ be a continuous, real-valued, symmetric approximate identity. Then $f \mapsto L\varphi_n(f)$ is an integral operator and hence compact, and symmetry and being real-valued implies that they are self-adjoint. Thus, we may diagonalize $L^2(G)$ against these operators. Each nonzero eigenspace is finite-dimensional and G -invariant (for right translations) and hence in $L^2(G)^{\text{fin}}$, so $\text{im } L\varphi_n \subseteq L^2(G)^{\text{fin}}$. But for any $f \in L^2(G)$, we have $L\varphi_n(f) \rightarrow f$ as $n \rightarrow \infty$, so $\sum_n \text{im } L\varphi_n$ is dense in $L^2(G)$! ■

Corollary 50. Let G be a compact group. For each irreducible finite-dimensional representation V , let $\chi_V \in C(G)$ be the character of V . Then $\{\chi_V\}_V$ is an orthonormal basis of the space $L^2(G)^G$ of conjugation-invariant functions.

Sketch. Certainly the functions $\{\chi_V\}_V$ are orthonormal by orthogonality of matrix coefficients. To get density, we use the Peter–Weyl theorem: we have a commutative diagram

$$\begin{array}{ccc} \bigoplus_V (V \otimes V^*)^G & \xrightarrow{\xi} & L^2(G)^G \\ \downarrow & & \downarrow \\ \bigoplus_V (V \otimes V^*) & \xrightarrow{\xi} & L^2(G) \end{array}$$

where each downward inclusion admits a section given by averaging via the integral. Thus, to approximate in the top-right, we push to the bottom-right, approximate by the bottom-left, and then apply the section to get back in the top-left. ■

2.5 Applications

Proposition 51. Any compact Lie group G admits a faithful finite-dimensional representation.

Sketch. Note that every element of G acts nontrivially on $L^2(G)$ and hence on $L^2(G)^{\text{fin}}$ by density. Thus,

$$\bigcap_V \ker \rho_V = \{1\},$$

where the intersection is over all finite-dimensional irreducible representations of G . Enumerating the intersection gives a descending sequence of closed Lie subgroups of G , which must eventually stabilize because G is a Lie group with finite component group, so some finite intersection equals $\{1\}$. ■

Proposition 52. Any compact group is an inverse limit of compact Lie groups.

Proof. Enumerate the finite-dimensional irreducible representations as $\{V_i\}_i$, and define $W_n := \bigoplus_{i \leq n} V_i$, and let K_n be the kernel of W_n . As in the previous proof, $\bigcap_n K_n = \{1\}$, but now $G/K_n \subseteq \text{U}(W_n)$ is some compact Lie group, so we are done upon writing $G = \lim_n G/K_n$. ■

Corollary 53. Let G be a compact group. Then $L^2(G)^{\text{fin}}$ is dense in $C(G)$, equipped with $\|\cdot\|_\infty$.

Proof. This follows from the Stone–Weierstrass theorem. The proposition is used to show that $L^2(G)^{\text{fin}}$ separates points. ■

Corollary 54. Let G be a compact Lie group, and let V be any faithful finite-dimensional representation of G . Then $V \oplus V^*$ $\otimes$ -generates $\text{Rep } G$.

Proof. Define $W := V \oplus V^*$; note that $\det W = 1$. Because $\text{Rep } G$ is semisimple, it is enough to realize all irreducible representations of G , so it is enough to realize $L^2(G)^{\text{fin}}$. In fact, we claim that the matrix coefficient map $\text{Sym}(W \otimes W^*) \rightarrow L^2(G)^{\text{fin}}$ is surjective: the generated subalgebra A is left-invariant, separates points (because W is faithful), and it is stable under complex conjugation because any $g \in G$ has $\rho_W(g)^\dagger = \rho_W(g)^{-1} = \rho_{\wedge^{\dim W - 1} W}(g)$. Thus, A is dense in $C(G)$, so we find $A = L^2(G)^{\text{fin}}$ by computing the multiplicity spaces $\text{Hom}_G(V, A)$ as before to be V^* . ■

3 Real Forms

3.1 Counting with Cocycles

Proposition 55. Fix a Galois field extension L/K and a Lie algebra $\mathfrak{g}$ over K . Then $H^1(\text{Gal}(L/K); \text{Aut } \mathfrak{g}_L)$ classifies Lie algebras $\mathfrak{g}'$ over K such that $\mathfrak{g}'_L \cong \mathfrak{g}_L$.

Sketch. Given a 1-cocycle $c: \text{Gal}(L/K) \rightarrow \text{Aut } \mathfrak{g}_L$, recover $\mathfrak{g}'$ by taking Galois-invariants. Conversely, given $\mathfrak{g}'$, define the 1-cocycle by sending $\sigma \in \text{Gal}(L/K)$ to the automorphism given by moving around the following square.

$$\begin{array}{ccc} \mathfrak{g}'_L & \xrightarrow{\sim} & \mathfrak{g}_L \\ \sigma \downarrow & & \downarrow \sigma \\ \mathfrak{g}'_L & \xrightarrow{\sim} & \mathfrak{g}_L \end{array}$$

Modifying the choice of isomorphism adjusts the 1-cocycle by a corresponding 1-coboundary. ■

Corollary 56. Fix a semisimple complex Lie algebra $\mathfrak{g}$. Then real forms of $\mathfrak{g}$, denoted $\mathfrak{g}_{(s)}$, are classified by $s \in \text{Aut } \mathfrak{g}$ satisfying $s\bar{s}$, up to the conjugation action $a \cdot s = as\bar{a}^{-1}$.

Proof. Apply the proposition to the split real form of $\mathfrak{g}$. In particular, define s to be the image of the nontrivial element of $\text{Gal}(\mathbb{C}/\mathbb{R})$ along the 1-cocycle. ■

Remark 57. By construction, $\mathfrak{g}_{(s)} = \{X \in \mathfrak{g}_{\mathbb{C}} : X = \overline{sX}\}$.

Definition 58 (inner class). Two real forms $\mathfrak{g}_{(s)}$ and $\mathfrak{g}_{(s')}$ are in the same *inner class* if and only if s and s' are in the same connected component of $\text{Aut } \mathfrak{g}$ (up to conjugation).

Definition 59 (quasisplit). The real form $\mathfrak{g}_{(s)}$ is *quasisplit* if and only if the component of s in G_{ad} is trivial (up to conjugation).

Definition 60 (compact form). The compact real form $\mathfrak{g}^c$ is defined $\mathfrak{g}_{(\tau)}$, where τ is the involution defined on the Serre presentation by

$$\begin{cases} \tau(h_i) = h_i, \\ \tau(e_i) = -f_i, \\ \tau(f_i) = -e_i. \end{cases}$$

Remark 61. The Killing form on $\mathfrak{g}^c$ is negative-definite: one can divide this calculation into $\mathfrak{h} \cap \mathfrak{g}^c$ (handled by direct calculation) and the pieces $(\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}) \cap \mathfrak{g}^c$ (handled by using the Weyl group to reduce to the case of simple roots and then to $\mathfrak{sl}_2$). Thus, the corresponding subgroup $G_{\text{ad}}^c \subseteq G_{\text{ad}}$ is compact.

Remark 62. A similar calculation on the root decomposition shows that the Killing form on the split real form has signature $(\text{rank } \mathfrak{g} + \#\Phi^+, \#\Phi^+)$.

Remark 63. The image of τ in $\text{Aut } D$ is $-w_0$, where w_0 is the long Weyl element. Indeed, τ acts by (-1) on the root system.

Proposition 64. Fix a semisimple complex Lie algebra $\mathfrak{g}$. Then real forms $\mathfrak{g}_\theta$ are classified by conjugacy classes of involutions θ commuting with $X \mapsto \tau \bar{X}$.

Sketch. Set $\omega := \bar{\tau}$ for brevity. Then other antilinear involutions take the form $\omega \circ g$ for some $g \in \text{Aut } \mathfrak{g}$. By replacing g with $\theta := g|g|$ (note $\theta = |g|^{-1/2} g \omega(|g|^{1/2})$ also), we may assume $\theta^2 = 1$. Similarly judiciously taking square roots shows that the conjugacy relations match up as well. ■

Definition 65. Let $\mathfrak{g}_\theta$ be a real form of $\mathfrak{g}$. Then we define $G_{\text{ad},\theta} := G_{\text{ad}}^{\omega_\theta}$, where $\omega_\theta := \bar{\tau} \circ \theta$ is the corresponding antilinear involution.

3.2 The Vogan Game

Remark 66. Given θ , let $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ be the eigenspace decomposition, where $\mathfrak{k} = \mathfrak{g}^{\theta=1}$ and $\mathfrak{p} = \mathfrak{g}^{\theta=-1}$. Then by construction, $\mathfrak{g}_\theta = \mathfrak{k}^c \oplus i\mathfrak{p}^c$. This is an orthogonal decomposition for the Killing form.

Example 67. The signature of the Killing form on $\mathfrak{g}_\theta$ is $(\dim \mathfrak{p}, \dim \mathfrak{k})$. It follows that the split real form has $\dim \mathfrak{k} = \#\Phi^+$.

Definition 68 (Vogan diagram). A Vogan diagram is a Dynkin diagram with the following decorations: each vertex is either

- connected with an auxiliary edge corresponding to an involution of the Dynkin diagram, or
- colored black or white.

Proposition 69. Each real form $\mathfrak{g}_\theta$ produces a Vogan diagram, and each Vogan diagram comes from a real form.

Sketch. By choosing a Cartan subalgebra $\mathfrak{h}_+$ of $\mathfrak{k}^c$ first and then extending it maximally by $\mathfrak{h}_-$ to $\mathfrak{ip}^c$, one can find a θ -stable Cartan subalgebra of $\mathfrak{g}^c$. It turns out that $\mathfrak{h}_-$ contains no coroots, so a generic $t \in \mathfrak{h}_+$ defines a polarization. Thus, θ now acts on the root decomposition preserving the positive roots and thus the simple roots. Thus, it induces an automorphism of the Dynkin diagram; by appropriate normalization, if $\theta(\alpha_i) \neq \alpha_i$, we may define $e_{\theta(i)} := \theta(e_i)$, and this corresponds to connected roots. Otherwise, $\theta(i) = i$, so $\theta(e_i) = \pm e_i$. The $+$ case corresponds to a white color, and the $-$ case corresponds to a black color. One can read the Vogan diagram backwards to recover θ . ■

Example 70. If $\mathfrak{g}_\theta$ is in the compact inner class, then θ must act by identity on the Dynkin diagram, so $\mathfrak{h} \subseteq \mathfrak{k}$, so $\text{rank } \mathfrak{k} = \text{rank } \mathfrak{g}$.

Remark 71 (Vogan game). Suppose that the vertex α_i is black. Then conjugating θ by s_i adjusts the sign of e_j from $\alpha_j(\theta)$ to $\alpha_j(s_i\theta)$, which is $s_i\alpha_j(\theta)$, which is

$$\alpha_j(\theta)\alpha_i(\theta)^{-a_{ij}}.$$

Thus, exactly the vertices α_j adjacent to α_i with a_{ij} (namely, pointing in the correct direction with odd multiplicity) change color.

Remark 72. If all vertices of a Vogan diagram (in the compact inner class) are white except for α_i , and ω_i is miniscule, then $\mathfrak{k}$ is the sum of $\mathfrak{gl}_1$ and the white vertices. Indeed, $\mathfrak{k}$ has all $\mathfrak{h}$, and it features the root spaces $\mathfrak{g}_\alpha$ which feature α_i an even number of times. This means that $(\alpha, \omega_i) = 0$ because ω_i is miniscule.

Remark 73. If all vertices of a Vogan diagram (in the compact inner class) are white except for α_i , and ω_i is maximal, then $\mathfrak{k}$ is the sum of $\mathfrak{sl}_2$ and the white vertices. Indeed, one argues as in the previous remark, using maximality to note that the only root α featuring α_i at least twice is ω_i .

Remark 74. If all vertices of the Vogan diagram are white, then $\mathfrak{k}$ is given by “folding” the Dynkin diagram in half (and adjusting lengths appropriately). Indeed, the new root system features the simple roots $\{\alpha_i : \theta(i) = i\}$ and $\{\alpha_i + \alpha_{\theta(i)} : \theta(i) \neq i\}$.

3.3 Polar Decomposition

Definition 75. Let $\mathfrak{g}_\theta$ be a real form of $\mathfrak{g}$. Then define K_θ^c to be the fixed points of $\omega_\theta := \omega \circ \theta$ in $G_{\text{ad},\theta}$.

Remark 76. One can expand out the definition to show that $\omega(g) = g^{-\dagger}$, where the implicit Hermitian form is the unique Hermitian extension of the Killing form restricted to $\mathfrak{g}^c$. Thus, $K_\theta^c = (G_{\text{ad}}^c)^{\omega_\theta}$ is subgroup of $G_{\text{ad},\theta}$ acting by unitary operators.

Definition 77. Let $\mathfrak{g}_\theta$ be a real form of $\mathfrak{g}$. Then define $P_\theta := \exp(\mathfrak{p}_\theta)$, where $\mathfrak{p}_\theta := \mathfrak{ip}^c$.

Remark 78. One can show that $X \in \mathfrak{g}$ acts by a Hermitian operator on $\mathfrak{g}$ if and only if $\omega(X) = -X$. Thus, $X \in \mathfrak{g}_\theta$ acts by a Hermitian operator if and only if $X \in \mathfrak{ip}^c$. Because the exponential is a diffeomorphism from Hermitian operators to positive Hermitian operators, we conclude that $P_\theta \subseteq G_{\text{ad},\theta}$ is the submanifold of positive Hermitian operators.

Theorem 79 (Polar decomposition). The multiplication map $K_\theta^c \times P_\theta \rightarrow G_{\text{ad},\theta}$ is a diffeomorphism.

Sketch. We construct an explicit smooth inverse. The idea is to do a polar decomposition in $\text{GL}(\mathfrak{g})$. Namely, the operator Ad_g is uniquely factored as $U_g R_g$, where $U_g \in \text{GL}(\mathfrak{g})$ is unitary and $R_g \in \text{GL}(\mathfrak{g})$ is positive Hermitian; we would like to show that $g \mapsto (U_g, R_g)$ is our inverse. Well, $R_g := (\text{Ad}_g^\dagger \text{Ad}_g)^{1/2}$, and an investigation of the action on the root decomposition reveals $R_g \in \text{Aut}(\mathfrak{g})$ and then $R_g \in P_\theta$. Thus, $U_g \in G_{\text{ad},\theta}$ and so $U_g \in K_\theta^c$. ■

4 Compact Groups

4.1 The Fundamental Group

Lemma 80. The group G_{ad} has trivial center.

Proof. Suppose $z \in G_{\text{ad}}$ is central. Then $\text{Ad}_z : \mathfrak{g} \rightarrow \mathfrak{g}$ is a scalar, so $z \in \text{GL}(\mathfrak{g})$ is a scalar matrix. However, this means $z[X, Y] = [zX, zY]$ for all $X, Y \in \mathfrak{g}$, so $z = 1$ follows. ■

Proposition 81. Let G be a complex Lie group with semisimple Lie algebra $\mathfrak{g}$ and center Z . For each dominant integral weight λ , let $\chi_\lambda : Z \rightarrow \mathbb{C}^\times$ be the central character of L_λ . Then $\chi_\bullet$ extends uniquely to an injective homomorphism

$$P/Q \rightarrow \text{Hom}(Z, \mathbb{C}^\times).$$

Sketch. To show that $\chi_\bullet$ extends to a homomorphism $P \rightarrow \text{Hom}(Z, \mathbb{C}^\times)$, note that $L_{\lambda+\mu} \subseteq L_\lambda \otimes L_\mu$, so $\chi_{\lambda+\mu} = \chi_\lambda \chi_\mu$ for dominant integral λ and μ . It remains to compute the kernel.

- If $\chi_\lambda = 1$, then L_λ descends to a representation of G/Z , which is G_{ad} . But $\mathfrak{g}$ is a faithful unimodular representation of G_{ad} , so $\mathfrak{g} \otimes$ -generates $\text{Rep } G_{\text{ad}}^c$. Thus, L_λ may only feature weights from Q , so $\lambda \in Q$.
- To show $Q \subseteq \ker \chi_\bullet$, it's enough to show that, for any root α , we have $\chi_\lambda \chi_\theta = \chi_{\lambda+\alpha}$ for large enough dominant integral weights λ . In fact, $L_{\lambda+\alpha} \subseteq L_\lambda \otimes \mathfrak{g}$ for λ large enough: by unwinding highest weight theory and the Serre presentation, we see that the space of maps $L_{\lambda+\alpha} \rightarrow \mathfrak{g} \otimes L_\lambda$ is given by

$$\mathfrak{g}[\alpha]_\lambda = \left\{ v \in \mathfrak{g}[\alpha] : f_i^{(\lambda, \alpha_i^\vee)} v = 0 \text{ for all } i \right\},$$

but the condition is automatically true if λ is large. ■

A Type A_n

- Lie algebra

- $\mathfrak{g} = \mathfrak{sl}(n+1)$.
- $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_{n+1}) : a_1 + \dots + a_{n+1} = 0\}$.
- $\mathfrak{h}^*$ is spanned by the projections e_i : $\text{diag}(a_1, \dots, a_{n+1}) \mapsto a_i$, where $e_1 + \dots + e_{n+1} = 0$.
- Killing form $K(X, Y) = 2(n+1) \text{tr } XY$.

- Root system

root	root space	coroot
$e_i - e_j$	E_{ij}	$E_{ii} - E_{jj}$

- Lattices

- Weight lattice $P = \{(\lambda_1, \dots, \lambda_{n+1}) \in \mathbb{R}^{n+1} / (1, \dots, 1)\mathbb{R} : \lambda_i - \lambda_{i+1} \in \mathbb{Z} \text{ for all } i\}$.
- Dominant weights $P_+ = \{(\lambda_1, \dots, \lambda_n) \in P : \lambda_1 \geq \dots \geq \lambda_n\}$.
- Root lattice $Q = \{(\lambda_1, \dots, \lambda_{n+1}) \in \mathbb{Z}^{n+1} : \lambda_1 + \dots + \lambda_{n+1} = 0\}$.
- $P/Q \cong \mathbb{Z}/(n+1)\mathbb{Z}$ by $(\lambda_1, \dots, \lambda_{n+1}) \mapsto \sum_i \lambda_i$.

- Polarization

- The positive roots are $\{e_i - e_j : i < j\}$.
- The simple roots are $\{e_i - e_{i+1} : i\}$. Set $\alpha_i := e_i - e_{i+1}$.
- Dynkin diagram:

$$\alpha_1 \text{ --- } \alpha_2 \text{ --- } \dots \text{ --- } \alpha_n$$

- The fundamental weights are $\{\omega_i\}_i$ where $\omega_i := (1, \dots, 1, 0, \dots, 0)$ has i ones.
- Maximal root $\theta = e_{n+1} - e_n$.

- Weyl group

- Weyl group is $W = S_{n+1}$ acting by permutations of coordinates.
- The simple reflection s_i is $(i, i+1)$.
- $w_0 = (n+1, n, \dots, 2, 1)$.

- Representation theory

- $\rho = (n/2, n/2 - 1, \dots, 1 - n/2, -n/2)$ and $\rho^\vee = (n/2, n/2 - 1, \dots, 1 - n/2, -n/2)$.
- The miniscule weights are $\{0\} \sqcup \{\omega_i : i\}$.
- Fundamental representations are $L_{\omega_i} = \wedge^i \mathbb{C}^{n+1}$.
- Exponents are $\{1, 2, \dots, n\}$.

- Group theory

- $G_{\text{sc}} = \text{SL}_{n+1}(\mathbb{C})$.
- $G_{\text{ad}} = \text{PSL}_{n+1}(\mathbb{C})$.

B Type B_n

- Lie algebra

- $\mathfrak{g} = \mathfrak{so}(2n+1)$, preserving

$$B = \begin{bmatrix} & & 1_n \\ & 1_n & \\ & & \\ & & & 1 \end{bmatrix}.$$

- $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n, 0) : a_1, \dots, a_n\}$.
- $\mathfrak{h}^*$ is spanned by the projections $e_i : \text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n, 0) \mapsto a_i$.
- Killing form $K(X, Y) = 2(n+1) \text{tr } XY$.

- Root system

root	length	root space	coroot
$e_i - e_j$	long	$E_{ij} - E_{i+n, j+n}$	$H_i - H_j$
$e_i + e_j$	long	$E_{i, j+n} - E_{j, i+n}$	$H_i + H_j$
$-e_i - e_j$	long	$E_{i+n, j} - E_{j+n, i}$	$-H_i - H_j$
e_i	short	$E_{i, 2n+1} - E_{2n+1, i+n}$	$2H_i$
$-e_i$	short	$E_{n+i, 2n+1} - E_{2n+1, i}$	$-2H_i$

Here, $H_i = E_{ii} - E_{i+n, i+n}$.

- Lattices

- Weight lattice $P = \{(\lambda_1, \dots, \lambda_n) \in (\frac{1}{2}\mathbb{Z})^n : \lambda_i - \lambda_{i+1} \in \mathbb{Z} \text{ for all } i\}$.
- Dominant weights $P_+ = \{(\lambda_1, \dots, \lambda_n) \in P : \lambda_1 \geq \dots \geq \lambda_n \geq 0\}$.
- Root lattice $Q = \mathbb{Z}^n$.
- $P/Q \cong \mathbb{Z}/2\mathbb{Z}$ by $(\lambda_1, \dots, \lambda_n) \mapsto \lambda_1$.

- Polarization

- The positive roots are $\{e_i - e_j : i < j\} \cup \{e_i + e_j : i < j\} \cup \{e_i : i\}$.
- The simple roots are $\{e_i - e_{i+1} : i\} \cup \{e_n\}$. Set $\alpha_i := e_i - e_{i+1}$ and $\alpha_n := e_n$.
- Dynkin diagram:

$$\alpha_1 \text{ --- } \alpha_2 \text{ --- } \dots \text{ --- } \alpha_{n-1} \implies \alpha_n$$

- The fundamental weights are $\{\omega_i\}_i$ where $\omega_i := (1, \dots, 1, 0, \dots, 0)$ has i ones for $i < n$ and $\omega_n := \frac{1}{2}(1, \dots, 1)$.
- Maximal root $\theta = e_1 + e_2$.

- Combinatorics

- Weyl group is $W = S_n \ltimes \{\pm 1\}^n$ acting by permutations of coordinates and sign changes.
- The simple reflection s_i is $(i, i+1)$ for $i < n$, and s_n is $(+1, \dots, +1, -1)$.
- $w_0 = (-1, \dots, -1)$.

- Representation theory

- $\rho = (n-1/2, n-3/2, \dots, 3/2, 1/2)$ and $\rho^\vee = (n, n-1, \dots, 2, 1)$.
- The miniscule weights are $\{0\} \sqcup \{\omega_1\}$.
- Fundamental representations are $L_{\omega_i} = \wedge^i \mathbb{C}^{2n+1}$ for $i < n$ and the standard Clifford module for $i = n$.
- Exponents are $\{1, 3, \dots, 2n-3, 2n-1\}$.

- Group theory

- $G_{\text{sc}} = \text{Spin}_{2n+1}(\mathbb{C})$.
- $G_{\text{ad}} = \text{SO}_{2n+1}(\mathbb{C})$.

C Type C_n

- Lie algebra

- $\mathfrak{g} = \mathfrak{sp}(2n)$, preserving

$$B = \begin{bmatrix} & -1_n \\ 1_n & \end{bmatrix}.$$

- $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n) : a_1, \dots, a_n\}$.
- $\mathfrak{h}^*$ is spanned by the projections $e_i : \text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n) \mapsto a_i$.
- Killing form $K(X, Y) = 2(n+1) \text{tr } XY$.

- Root system

root	length	root space	coroot
$e_i - e_j$	short	$E_{ij} - E_{i+n, j+n}$	$H_i - H_j$
$e_i + e_j$	short	$E_{i, j+n} - E_{j, i+n}$	$H_i + H_j$
$-e_i - e_j$	short	$E_{i+n, j} - E_{j+n, i}$	$-H_i - H_j$
$2e_i$	long	$E_{i, i+n}$	H_i
$-2e_i$	long	$E_{i+n, i}$	$-H_i$

Here, $H_i = E_{ii} - E_{i+n, i+n}$.

- Lattices

- Weight lattice $P = \mathbb{Z}^n$.
- Dominant weights $P_+ = \{(\lambda_1, \dots, \lambda_n) \in P : \lambda_1 \geq \dots \geq \lambda_n \geq 0\}$.
- Root lattice $Q = \{(\lambda_1, \dots, \lambda_n) \in \mathbb{Z}^n : \lambda_1 + \dots + \lambda_n \equiv 0 \pmod{2}\}$.
- $P/Q \cong \mathbb{Z}/2\mathbb{Z}$ by $(\lambda_1, \dots, \lambda_n) \mapsto \sum_i \lambda_i$.

- Polarization

- The positive roots are $\{e_i - e_j : i < j\} \cup \{e_i + e_j : i < j\} \cup \{2e_i : i\}$.
- The simple roots are $\{e_i - e_{i+1} : i\} \cup \{2e_n\}$. Set $\alpha_i := e_i - e_{i+1}$ and $\alpha_n := 2e_n$.
- Dynkin diagram:

$$\alpha_1 \text{ --- } \alpha_2 \text{ --- } \dots \text{ --- } \alpha_{n-1} \Longleftarrow \alpha_n$$

- The fundamental weights are $\{\omega_i\}_i$ where $\omega_i := (1, \dots, 1, 0, \dots, 0)$ has i ones.
- Maximal root $\theta = 2e_1$.

- Combinatorics

- Weyl group is $W = S_n \ltimes \{\pm 1\}^n$ acting by permutations of coordinates and sign changes.
- The simple reflection s_i is $(i, i+1)$ for $i < n$, and s_n is $(+1, \dots, +1, -1)$.
- $w_0 = (-1, \dots, -1)$.

- Representation theory

- $\rho = (n, n-1, \dots, 2, 1)$ and $\rho^\vee = (n-1/2, n-3/2, \dots, 3/2, 1/2)$.
- The miniscule weights are $\{0\} \sqcup \{\omega_1\}$.
- Fundamental representations are $L_{\omega_i} = \ker(\iota_B : \wedge^i \mathbb{C}^{2n} \rightarrow \wedge^{i-2} \mathbb{C}^{2n})$, where ι_B is contraction:

$$\iota_B(v_1 \wedge \dots \wedge v_i) := \sum_{r < s} (-1)^{r+s+1} B(v_r \wedge v_s) (v_1 \wedge \dots \wedge \widehat{v}_r \wedge \dots \wedge \widehat{v}_s \wedge \dots \wedge v_i).$$

- Exponents are $\{1, 3, \dots, 2n-3, 2n-1\}$.

- Group theory

- $G_{\text{sc}} = \text{Sp}_{2n}(\mathbb{C})$.
- $G_{\text{ad}} = \text{Sp}_{2n}(\mathbb{C})/\{\pm 1\}$.

D Type D_n

- Lie algebra

- $\mathfrak{g} = \mathfrak{so}(2n)$, preserving

$$B = \begin{bmatrix} & 1_n \\ 1_n & \end{bmatrix}.$$

- $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n) : a_1, \dots, a_n\}$.
- $\mathfrak{h}^*$ is spanned by the projections $e_i : \text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n) \mapsto a_i$.
- Killing form $K(X, Y) = 2(n+1) \text{tr } XY$.

- Root system

root	root space	coroot
$e_i - e_j$	$E_{ij} - E_{j+n, i+n}$	$H_i - H_j$
$e_i + e_j$	$E_{i, j+n} - E_{j, i+n}$	$H_i + H_j$
$-e_i - e_j$	$E_{i+n, j} - E_{j+n, i}$	$-H_i - H_j$

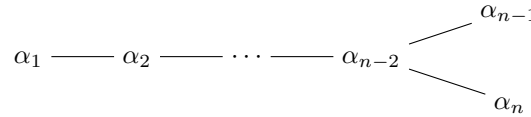
Here, $H_i = E_{ii} - E_{i+n, i+n}$.

- Lattices

- Weight lattice $P = \{(\lambda_1, \dots, \lambda_n) \in (\frac{1}{2}\mathbb{Z})^n : \lambda_i - \lambda_{i+1} \in \mathbb{Z} \text{ for all } i\}$.
- Dominant weights $P_+ = \{(\lambda_1, \dots, \lambda_n) \in P : \lambda_1 \geq \dots \geq \lambda_n, \lambda_{n-1} + \lambda_n \geq 0\}$.
- Root lattice $P = \{(\lambda_1, \dots, \lambda_n) \in \mathbb{Z}^n : \lambda_1 + \dots + \lambda_n \equiv 0 \pmod{2}\}$.
- $P/Q \cong (\mathbb{Z}/2\mathbb{Z})^2$ if n even and $P/Q \cong \mathbb{Z}/4\mathbb{Z}$ if n odd. It is generated by $(1, 0, \dots, 0)$ and $\frac{1}{2}(1, \dots, 1)$.

- Polarization

- The positive roots are $\{e_i - e_j : i < j\} \cup \{e_i + e_j : i < j\}$.
- The simple roots are $\{e_i - e_{i+1} : i\} \cup \{e_{n-1} + e_n\}$. Set $\alpha_i := e_i - e_{i+1}$ and $\alpha_n := e_{n-1} + e_n$.
- Dynkin diagram:



- The fundamental weights are $\{\omega_i\}_i$ where $\omega_i := (1, \dots, 1, 0, \dots, 0)$ has i ones for $i < n-1$ and $\omega_{n-1} := \frac{1}{2}(1, \dots, 1, -1)$ and $\omega_n := \frac{1}{2}(1, \dots, 1)$.
- Maximal root $\theta = e_1 + e_2$.

- Combinatorics

- Weyl group is $W = S_n \ltimes \{\pm 1\}_+^n$ acting by permutations of coordinates and even sign changes.
- The simple reflection s_i is $(i, i+1)$ for $i < n$, and s_n is $(+1, \dots, +1, -1, -1)$.
- If n even, $w_0 = (-1, \dots, -1)$ acts by (-1) on P/Q . If n odd, $w_0 = (-1, \dots, -1, +1)$ does not.

- Representation theory

- $\rho = (n-1, n-2, \dots, 1, 0)$ and $\rho^\vee = (n-1, n-2, \dots, 1, 0)$.
- The miniscule weights are $\{0\} \sqcup \{\omega_1, \omega_{n-1}, \omega_n\}$.
- Fundamental representations are $L_{\omega_i} = \wedge^i \mathbb{C}^{2n+1}$ for $i < n$ and the degree pieces of the standard Clifford module.
- Exponents are $\{1, 3, \dots, 2n-3, 2n-1\} \sqcup \{n\}$.

- Group theory

- $G_{\text{sc}} = \text{Spin}_{2n}(\mathbb{C})$.
- $G_{\text{ad}} = \text{SO}_{2n}(\mathbb{C})/\{\pm 1\}$.